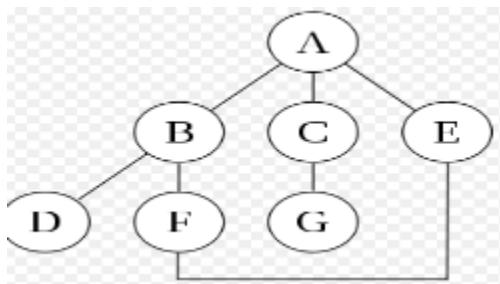


PART-A

1. Explain the difference between Symbolic AI and Sub-symbolic AI.
2. State two applications of AI in Healthcare and Finance.
3. Formulate a simple problem statement for an AI agent solving the 8-puzzle problem. Mention initial state and goal state.
4. Give the order of traversal for BFS and DFS for the following search tree. Avoid repeated states while expanding the nodes.



5. Define the Turing Test
6. Analyze why Breadth First Search is considered a complete search algorithm and under what conditions this property holds.
7. Differentiate between Propositional Logic and First-Order Logic.
8. Explain the concept of a Semantic Network in knowledge representation.
9. **Compare Semantic Networks, Frames, and Ontologies as knowledge representation techniques in AI.**
10. Compare AI and Machine Learning.
11. Define heuristic.
12. Compare BFS and DFS algorithms,
13. List the three levels of abstraction for modelling a knowledge based agent
14. Explain the role of Minimax Algorithm and how Alpha-Beta Pruning improves its efficiency.
15. Formulate a real-world problem as a Constraint Satisfaction Problem by clearly identifying the variables, domains, and constraints.

- 16 **Differentiate between Propositional Logic and First-Order Logic.**
- 17 **Represent a simple real-world statement (e.g., “All humans are mortal”) using First-Order Logic.**
- 18 Explain the importance of logic in Knowledge Representation in AI.
- 19 What is meant by AI ethics?
- 20 What is adversarial search?
- 21 What are frames in knowledge representation?

PART-B

- 1 Illustrate the steps involved in formulating problems with Example.
- 2 Infer the impact of Artificial Intelligence applications in Healthcare, Finance, and Robotics. Discuss their benefits, practical challenges, and ethical concerns, supporting your answer with relevant examples.
- 3 **Given an initial state of 8 puzzle problem initial state and Final State.**

2	8	3
1	6	4
7		5

1	2	3
8		4
7	6	5

Initial State

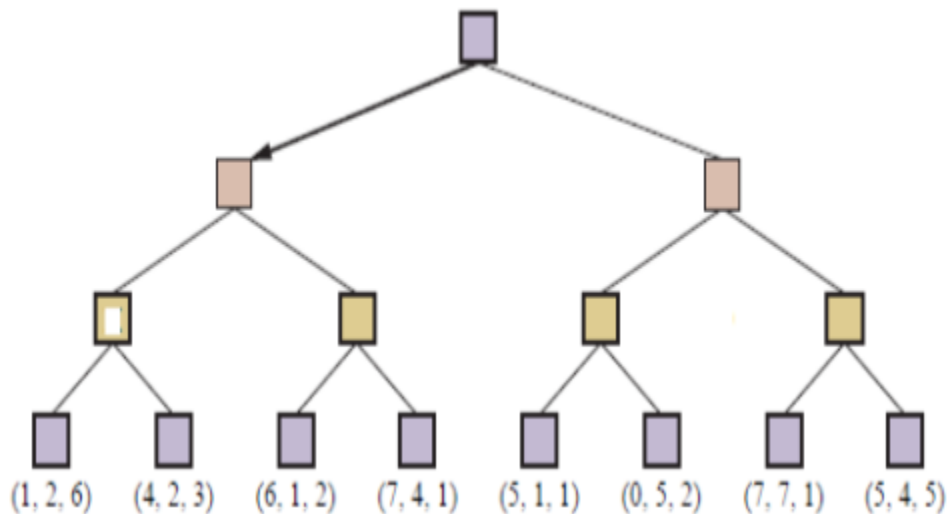
Final State

Find the most cost-effective path to reach the final state from initial state using A* algorithm.

- 4 Analyze and compare Symbolic AI and Sub-symbolic AI in terms of knowledge representation, learning mechanisms, reasoning capability, interpretability, and real-world applicability. Illustrate your answer with suitable examples.

5. Explain the **Turing Test** with diagram and example.

5 **Solve the three player game by Min-max algorithm method**



6 Apply the concept of constraint satisfaction problems by specifying the variables, domains and constraints and illustrate the map colouring problem with four regions and given adjacencies.

7 Classify the components of an ontology with examples in artificial intelligence.

8 Model a simple domain (e.g., library management or hospital system) using an Ontology and illustrate how it supports structured knowledge representation and reasoning.

9 Analyze the history and evolution of Artificial Intelligence

10 Examine the major challenges in ensuring data privacy and security within AI systems.

11. Explain alpha-beta pruning algorithm and the Minmax game playing algorithm with example?

12. With the heuristic information given below show the expansion of nodes in A* to obtain the Goal. "A" is the initial node and "L" is the goal node.

Nodes $h(n)$

A 8.0

B 6.3

C 6.3

D 5.0

E 4.0

F 5.1

G 5.0

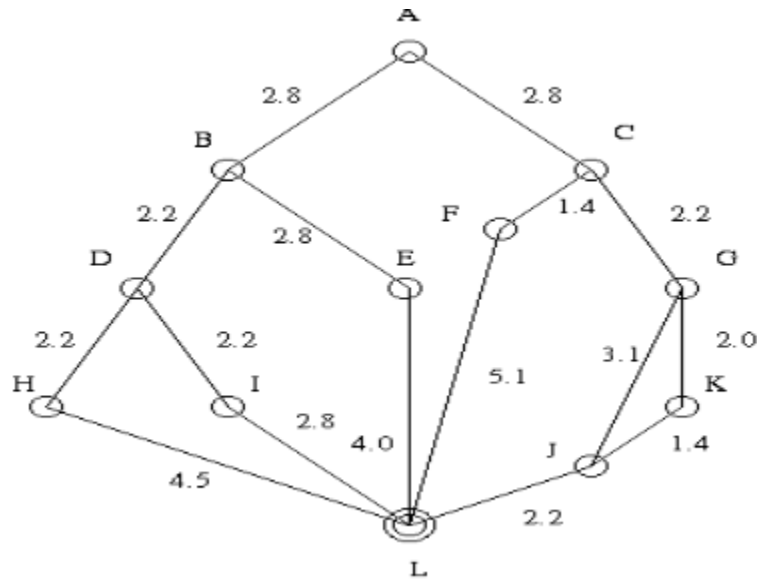
H 4.5

I 2.8

J 2.2

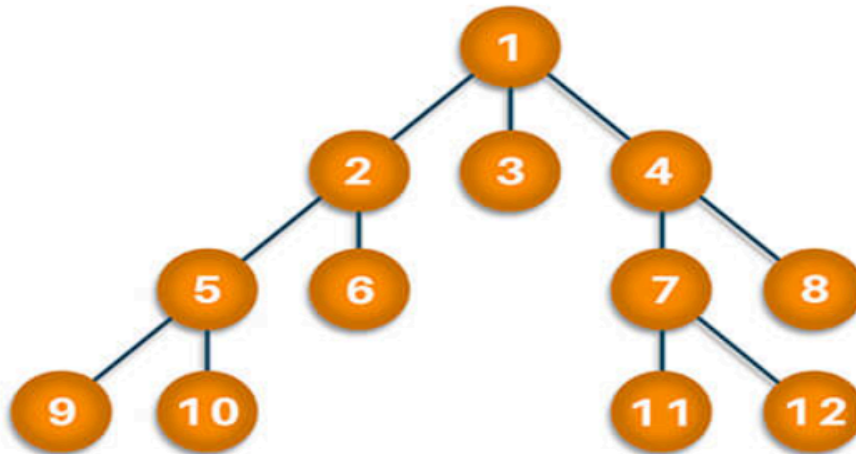
K 3.6

L 0.0



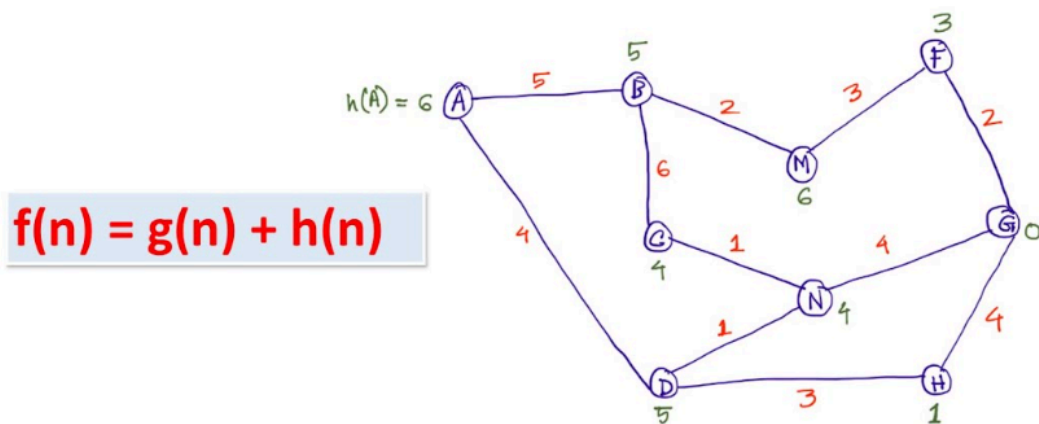
- 13 Analyze different knowledge representation techniques in AI, including Semantic Networks, Frames, and Ontologies

14. Apply DFS algorithm for given tree



- 14 Consider the following sentences: John likes all kinds of food; Apples are food; Chicken is food; Anything anyone eats and isn't killed by is food; Bill eats peanuts and still is alive; Sue eats everything Bill eats.
- (i) Translate these sentences into formulas in predicate logic.
- (ii) Convert the formulas of part an into clause form.

16. Solve the A* algorithm for given graph



17. Solve the Alpha beta pruning algorithm in details

